DOCTORAL APPLICATION TECHNICAL PORTFOLIO

Probabilistic Electricity Price Forecasting under Regime Shift

Static and adaptive conformal calibration in a leakage-controlled simulated day-ahead benchmark

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Abstract

This project asks whether prediction intervals can respond when electricity-price behaviour changes. It uses a simulated hourly benchmark and keeps model fitting, calibration and evaluation in time order. A seasonal-naive forecast is the point baseline. Seven pooled gradient-boosting models, one for each quantile level and all conditioned on forecast horizon, estimate marginal uncertainty for 24 hourly targets. Raw intervals are compared with pooled static conformalized quantile regression (CQR), horizon-specific static CQR and adaptive CQR. In the supplied Python 3.12.14 reference run, raw 80% intervals covered 54.8% of the final chronological evaluation observations. Adaptive CQR reached 82.6% coverage with a mean width of 22.50 EUR/MWh and an interval score of 30.95, but it still missed many extreme prices. I independently reran the pipeline on Windows with Python 3.13.15, passed all six tests, reproduced the same qualitative findings and then completed a pre-planned gamma 0.030 sensitivity experiment. All values are simulated benchmark results, not employer or live-market evidence.

Data and contribution boundary. All results come from the included simulator. The project applies established forecasting and conformal methods to build, test and question a reproducible research-training prototype. It is not presented as a new algorithm, a deployed trading system or evidence from a real electricity market.

1 Motivation and research question

My motivation came from work in the production-operations department of Tarim Oilfield's new-energy business. I compared photovoltaic forecast curves with actual output and saw that a gap could have several possible causes: weather, equipment availability, curtailment, maintenance or missing data. That experience made uncertainty and data quality concrete for me, but it did not provide data for this portfolio. The research question here is narrower: can an uncertainty interval adapt after a change without becoming so wide that it stops being useful?

Choices behind the experiment

- The rows are not randomly shuffled. A random split would allow later regimes to influence an earlier-looking evaluation and would not represent day-ahead use.
- The comparison begins with a seasonal-naive forecast and an interpretable tree-based quantile model. This separates centre-forecast accuracy from interval reliability before more complex models are considered.
- The main claims are deliberately limited to this simulated benchmark. Failure cases are reported alongside headline metrics.

2 Data and methods

2.1 Transparent simulated market

The simulator creates 10,080 hourly observations in three periods: stable operation, higher volatility and a renewable shift. Price depends on net load, scarcity, renewable surplus and changing noise. Load, renewable and temperature forecast proxies are created by perturbing simulated realised values with noise. This creates a controlled issue-time benchmark, although the proxies may be more informative than real operational forecasts. Short renewable outages and missing historical prices are also simulated. Missing prices are flagged and forward-filled; the pipeline stops rather than use future values if the required history begins with missing data. The true regime label is used only for evaluation.

Component	Implementation
Forecast setup	Seven pooled models, one per quantile, conditioned on horizon; 24 hourly targets
Past-only features	24-hour and 168-hour price lags, trailing moments and missingness rate
Future-known proxies	Day-ahead load, renewable and temperature forecasts
Target	Hourly price in EUR/MWh

2.2 Models and calibration

Baseline. The seasonal-naive forecast uses the price at the same target hour one day earlier.

Quantile model. One pooled histogram gradient-boosting model is fitted at each of seven quantile levels from q05 to q95. Horizon is an input feature; the implementation does not fit 24 separate models. Estimated quantiles are sorted within each row if they cross.

Static CQR comparators. The pooled version uses one fixed correction for all 24 horizons. A second static version learns a separate fixed correction for each horizon. This separates horizon conditioning from online adaptation.

Adaptive CQR. Each forecast horizon uses its own recent error buffer and a changing local miscoverage rate. The interval is calculated before the relevant outcome is added, so an observation cannot change its own interval.

2.3 Validation and metrics

The final 83 issue days form a chronologically separated evaluation block and are not used for model fitting or initial conformal calibration. The model is also refitted at three earlier origins, with calibration based only on past data. Each origin evaluates raw, pooled static, horizon-specific static and adaptive intervals. MAE checks the median forecast; coverage, width, pinball loss, approximate CRPS and interval score provide complementary checks. Because the final block is reused for post-hoc parameter sensitivity, it is described as an evaluation period rather than an untouched test set after analysis.

2.4 Relationship to existing work

Prior work establishes conformalized quantile regression, adaptive conformal inference, sequential methods for time series and electricity-price applications (Romano et al., 2019; Gibbs and Candes, 2021; Zaffran et al., 2022; Xu and Xie, 2023; O'Connor et al., 2025). This portfolio therefore contributes implementation, controlled comparisons and failure analysis, not a first application or new foundational method.

3 Supplied reference results

Simulated-result notice. The table and Figures 1-2 and 4 use the supplied Python 3.12.14 reference; Figure 3 uses my Windows Python 3.13.15 runs. None is a Tarim Oilfield, employer or live-market result.

Interval	Coverage	Width	Interval score	Median MAE
Raw quantiles	0.548	11.76	34.35	6.30
Static CQR pooled	0.924	27.64	32.18	6.30
Static CQR by horizon	0.933	29.29	33.56	6.30
Adaptive CQR	0.826	22.50	30.95	6.30

The median model lowers MAE from 11.27 to 6.30 EUR/MWh. Raw intervals are too narrow after the shift. Static CQR raises coverage but produces wider intervals. Adaptive CQR has the lowest observed interval score among the four methods in this one simulation, but that comparison does not establish general superiority in real markets.

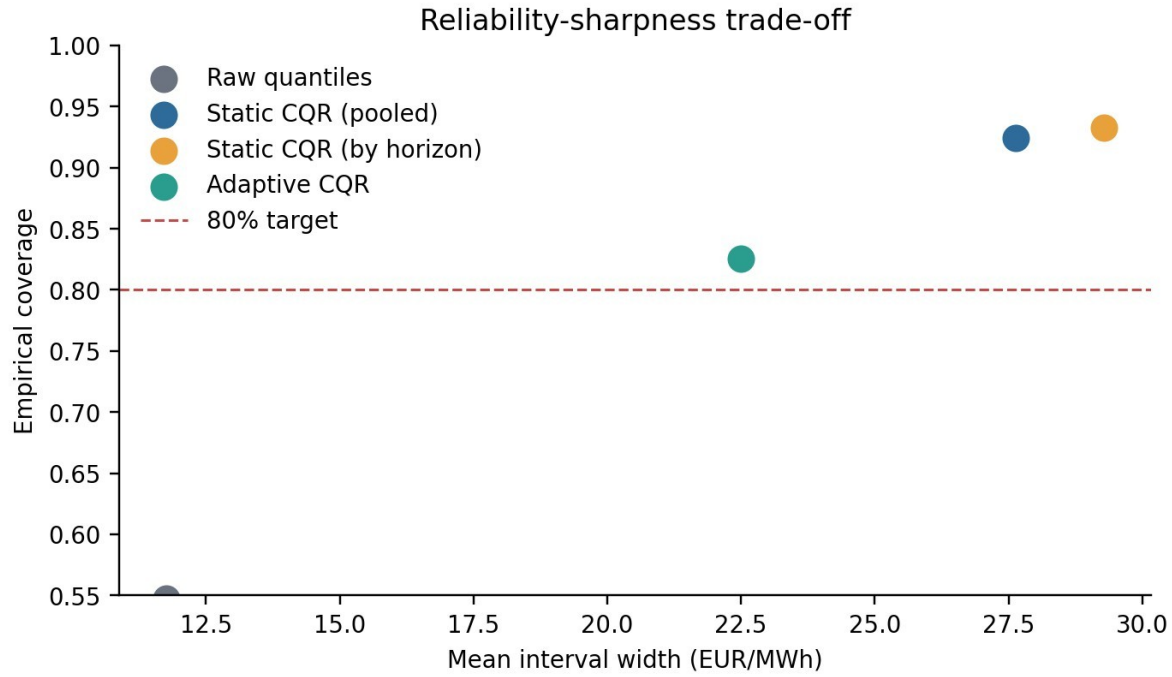


Figure 1 Reliability and sharpness in the chronological evaluation period; the dashed line is the 80% target.

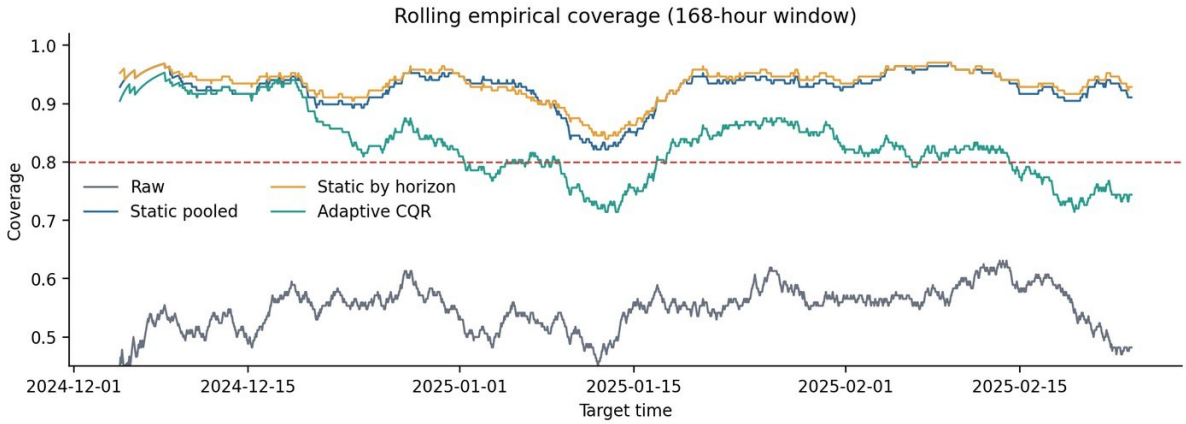


Figure 2 Rolling 168-hour coverage after the simulated renewable-shift boundary.

4 Applicant verification and additional experiment

Provenance and verification boundary. The repository began with an imported reference snapshot that predates its Git history. My dated commits record the work I personally completed after import: creating the local environment, passing the six unit tests, reproducing the baseline on Windows with Python 3.13.15, rerunning the supplied gamma 0.050 sensitivity and planning and running a new gamma 0.030 experiment. This report does not use those later commits to imply that I personally performed every earlier development step.

4.1 Independent Windows reproduction

On 25 September 2026, I created a local virtual environment with Python 3.13.15, installed the locked package versions and passed all six unit tests. I then reran the baseline with seed 20260817. The regenerated data had the same shape and non-numeric values as the supplied reference data. The maximum absolute numeric difference was 2.84 x 10^-14, and all numeric values matched after rounding to 10 decimal places. Because of those tiny floating-point differences, the file hash changed from 6bacc0ff... to 6d446b55...; the files are numerically equivalent for this comparison, not byte-identical.

Method	Reference coverage	Windows coverage	Reference width	Windows width
Raw quantiles	0.547691	0.556225	11.755545	11.896091
Static CQR pooled	0.924197	0.924197	27.640967	27.600691
Static CQR by horizon	0.932731	0.927711	29.289561	29.282753
Adaptive CQR	0.825803	0.826305	22.495669	22.362592

The Windows run reproduced the same practical conclusions: raw intervals under-covered, static calibration over-covered with wider intervals, and adaptive CQR sat closer to the 80% target with narrower intervals than the static alternatives. The small numeric changes are consistent with a different Python environment and regenerated floating-point data; they are documented rather than hidden.

4.2 Pre-planned gamma 0.030 experiment

Before running the experiment on 26 September 2026, I committed a short plan predicting that gamma 0.030 would produce results between the existing 0.015 and 0.050 settings. The results followed that pattern. All three rows below come from my local Windows environment and reuse the same simulated evaluation block.

Adaptive gamma	Coverage	Mean width	Interval score
0.015	0.826305	22.362592	30.906253
0.030	0.811245	21.671771	30.906484
0.050	0.804217	21.348917	31.116844

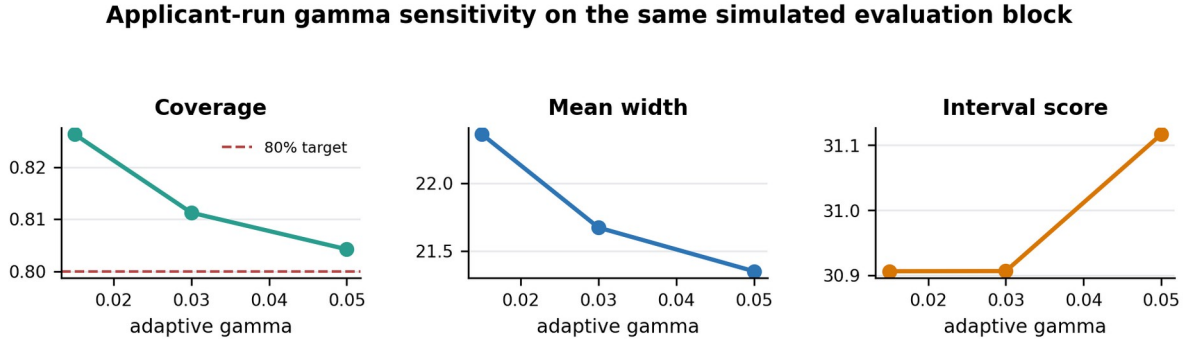


Figure 3 Applicant-run gamma sensitivity. The same evaluation block was reused, so the chart is diagnostic rather than a clean tuning study.

Gamma 0.030 reduced mean width by about 3.1% from 0.015 and moved coverage toward the 80% target, with a near-unchanged interval score (+0.00023). This is one simulated-run trade-off, not a universal optimum.

5 Stress tests, limitations and next steps

5.1 What failed

In the supplied reference run, adaptive-CQR coverage is 0.826 overall, 0.627 for the price-extreme subset and 0.710 for the feature-distance out-of-distribution flag. The method reacts to an error signal but cannot create information about rare spikes. Across the three earlier rolling-origin folds, adaptive coverage is 0.832, 0.747 and 0.839, so the middle fold remains difficult.

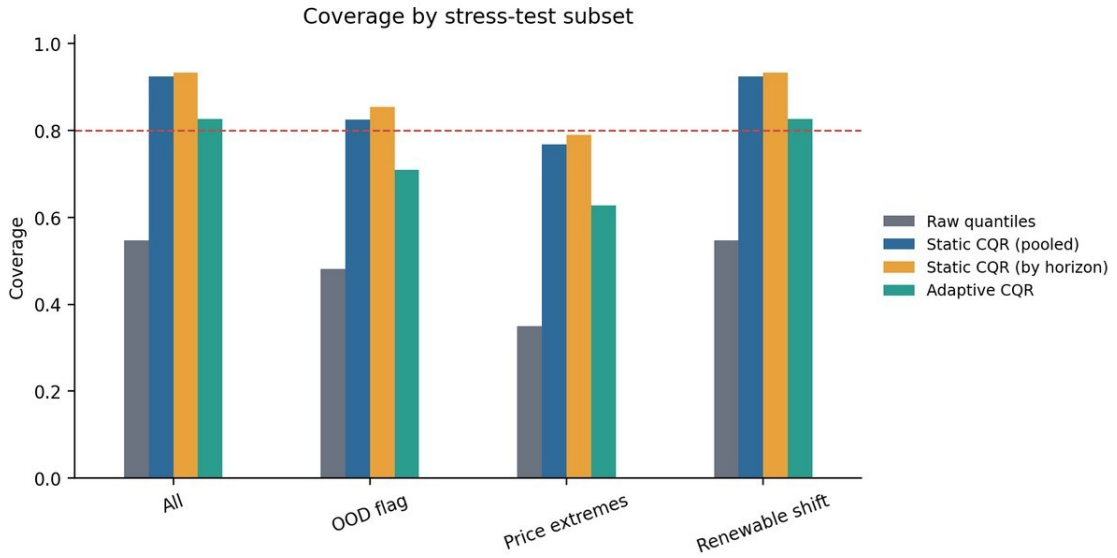


Figure 4 Calibration improves overall reliability in the simulated benchmark, but extreme prices remain a failure mode.

5.2 Main limitations in plain language

- One fixed simulation seed cannot show how much results vary across different simulated histories.
- Synthetic forecast proxies may be cleaner and more informative than real publication-time forecasts.
- Coverage averaged across all horizons can hide failures at particular hours.
- Marginal hourly quantiles do not automatically form reliable joint 24-hour price paths.
- Gamma 0.030 and 0.050 were examined on the same evaluation block, so neither may be presented as a winner selected on independent data.
- The prototype does not establish exchangeable-data guarantees under arbitrary temporal dependence or demonstrate profitable real-market operation.

5.3 Reproduction route

A clean rerun should write to a new folder so that committed evidence is not overwritten:

```
python scripts/run_experiment.py --config configs/base.yaml --output-root local_runs/baseline
```

The supplied reference manifest in outputs/run_manifest.json records the 6bacc0ff... fingerprint. My Windows reproduction manifest in evidence/runs/baseline/outputs/run_manifest.json records 6d446b55.... Exact package versions are in requirements-lock.txt, while evidence/README.md indexes the dated logs, results and personal experiment records.

5.4 Next research steps

The next defensible steps are to repeat the simulation across several seeds, report uncertainty using day-level blocks, inspect calibration by horizon and test genuine publication-time-safe data from sources such as ENTSO-E, Elexon or Open Power System Data. A later companion study can then examine whether better-calibrated forecast distributions improve constrained storage decisions, including behaviour near state-of-charge boundaries and under tail-risk objectives.

Responsible-use note. No employer, Tarim Oilfield or live-market data are included. Any later use of operational data would require permission, de-identification and a separate data-governance review.

References

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Repository evidence map

Item	Location
Project scope and quick start	README.md
Supplied and applicant reproduction record	REPRODUCIBILITY_RECORD.md
Evidence index	evidence/README.md
Personal baseline record	evidence/learning_log/baseline_reproduction.md
Pre-run plan and gamma 0.030 results	evidence/learning_log/gamma_030_plan.md and gamma_030_results.md
Project-history boundary	PROJECT_HISTORY.md